

Mathematics Stage 4

Rounding Decimals

Book 1 Rounding Decimals
Terminating and Recurring Decimals

Version: 261002

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Syllabus Content

MA4-FRC-C-01 represents and operates with fractions, decimals and percentages to solve problems

Round decimals to a specified degree of accuracy using approximations

- Round decimals to a given number of decimal places
- Apply the notation $\approx$ as a symbol of numerical approximation
- Reason why an approximation may be more appropriate than an exact answer and vice versa

Identify terminating and recurring decimals

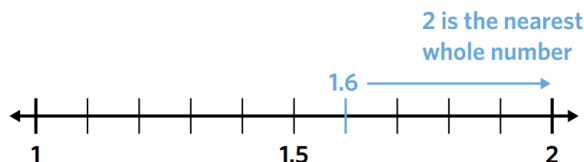
- Use either dot or vinculum notation for recurring (repeating) decimals
- Classify decimals as recurring or terminating

Rounding Decimals using a Number Line

① Rounding

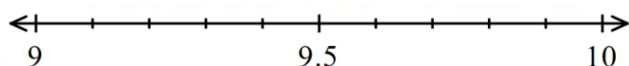
Rounding means to make a number simpler while keeping its value close to what it was. When rounding, we go to the nearest given place value.

E.g., when rounding 1.6 to the nearest whole, round to 2 as it is closer to 2.

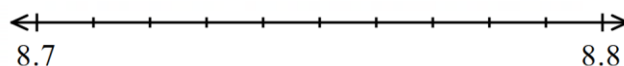


② Round decimals using a number line

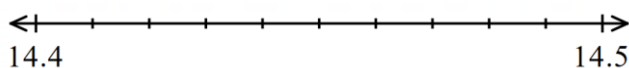
a Round 9.9 to the nearest whole



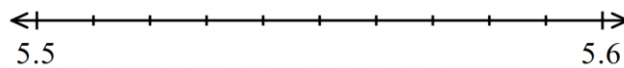
b Round 8.74 to the nearest tenth.



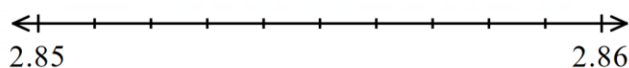
c Round 14.48 to the nearest tenth.



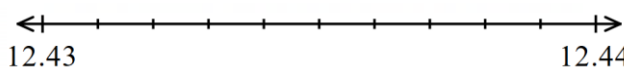
d Round 5.55 to the nearest tenth.



e Round 2.851 to the nearest hundredth.



f Round 12.436 to the nearest hundredth.

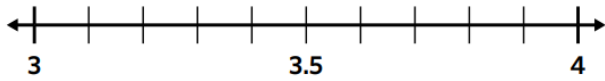


💡 Key Ideas

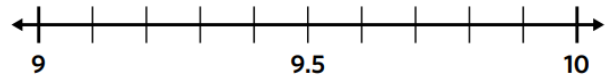
1. Rounding makes a number while keeping its value close to what it was.
2. We round when the number is halfway or more to the next place value.
3. We round down when a number is than halfway to the next place value.

1 Round these decimals to the nearest whole using a number line.

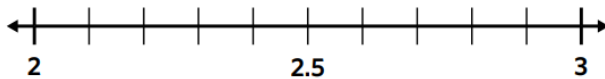
a $3.2 \approx \boxed{}$



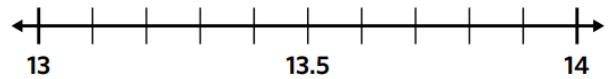
b $9.9 \approx \boxed{}$



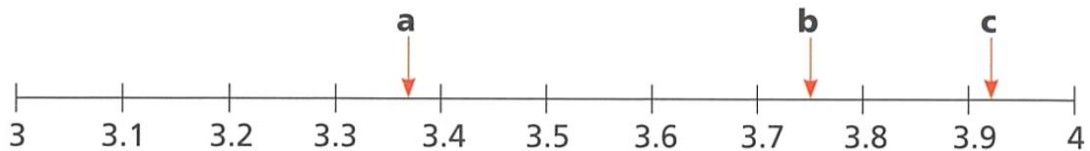
c $2.72 \approx \boxed{}$



d $13.46 \approx \boxed{}$

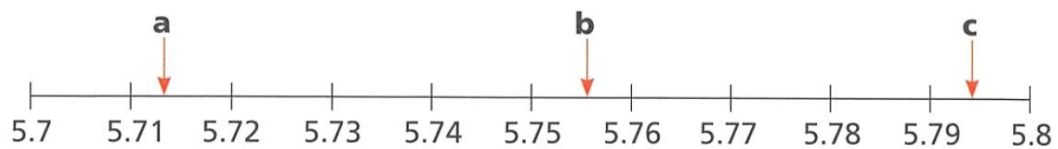


2 Use the number line to round each number to 1 decimal place



A: 3.4
B: 3.8
C: 3.9

3 Use the number line to round each number to 2 decimal places.



A: 5.71
B: 5.76
C: 5.79

Rounding using the Critical Digit Method

① The Critical Digit Method for Rounding

1. Draw a “cut” after the required place value.
2. Circle the **critical digit** right after the cut.
3. Decide to round up or down.
 - i. If critical digit is less than 5, round down.
 - ii. If critical digit is 5 or more, round up.
4. Round.
 - i. rounding down: Remove digits after the cut.
 - ii. rounding up: Add 1 to digit before the cut, then remove digits after the cut.

🧮 Example Round decimals to a place value

Round 22.53486 to the nearest hundredth.

1. Draw cut 2 2 . 5 3 | 4 8 6
 2. Critical digit 2 2 . 5 3 | 4 8 6
 3. Decide Round down
 4. Round 2 2 . 5 3
- $22.53486 \approx 22.53$

We draw a cut the required place value.

The critical digit is immediately to the of the cut.

When the critical digit is or less, we round down

✍ Guided Practice

a Round 0.954 to the nearest whole.
2 | 9 5 4

b Round 3.276 to the nearest hundredth.
3 . 2 7 | 6

c Round 9.871 to the nearest tenth.
9 . 8 7 1

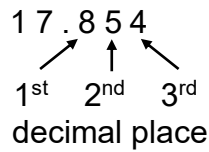
d Round 14.2129 to the nearest hundredth.
1 4 . 2 1 2 9

e Round 8.991 to the nearest tenth.

f Round 1.34294 to the nearest thousandth.

Decimal Places

Each digit to the right of the decimal point is called a decimal place.



Example Round decimals to a decimal place

Round 246.78 to one decimal place.

1. Draw cut 2 4 6 . 7 | 8

2. Critical digit 2 4 6 . 7 (8)

3. Decide Round up

4. Round 2 4 6 . 8

246.78 $\approx$ 246.8

The first decimal place is theths place value.

When the critical digit is or more, we round up.

Guided Practice

a Round 5.5323 to one decimal place.

5 . 5 | 3 2 3

b Round 3.276 to two decimal places.

3 . 2 7 6

c Round 9.8742 to two decimal places.

d Round 12.895 to two decimal places.

e Round 81.6814 to two decimal places.

f Round 241.9026343 to one decimal place.

Key Ideas

1. For decimals, you can round to a place value or to a d..... p.....
2. If the digit to the right of the rounding place is or more, round up. If it is or less, round down.

- 1 State the critical digit in each of the following numbers and decide whether to round up or down.

Question	Critical digit	Round up/down
25.8①74 rounded to 1 decimal place.	1	Down
25.81⑦4 rounded to 2 decimal places.	7	Up
25.817④ rounded to 3 decimal places	4	Down
25.⑧174 rounded to the nearest whole number.	8	Up

- 2 Round each of the following to the nearest whole number.
Draw a line where the number must be cut, circle the critical digit, then round.

a 27.⑥12	b 9.458	c 12.299	d 123.72
28	9	12	124
e 22.26	f 117.555	g 2.6132	h 10.7532
22	118	3	11

- 3 Round each of the following amounts to the nearest dollar.

a \$12.85	b \$30.50	c \$7.10	d \$1566.80
\$13	\$31	\$7	\$1567
e \$120.45	f \$9.55	g \$1.39	h \$36.19
\$120	\$10	\$1	\$36

- 4 The following decimals need to be rounded to 2 decimal places.

a 12.64⑤3	b 4.81932	c 157.281	d 465.3841
12.65	4.82	157.28	465.38
e 0.06031	f 203.5791	g 66.6666	h 7.995123
0.06	203.58	66.67	8.00

- 5 Round each of the following to 1 decimal place.

a 14.82	b 7.38	c 15.62	d 0.87
14.8	7.4	15.6	0.9
e 6.85	f 9.94	g 55.55	h 7.98
6.9	9.9	55.6	8.0

6 Explain the mistakes that these students have made, and give the correct answers.

a

6.2:34 34 is greater than 5, so $6.234 = 6.3$ to 1 decimal place

We only look at the digit immediately after the cut; 6.2

b

4.3:2 2 is less than 5, so $4.32 = 4.30$ to 1 decimal place

When we round down we don't change the digit before the cut; 4.3

c

8.9:7 7 is more than 5, so $8.97 = 8.10$ to 1 decimal place

When we round up from 9, it becomes 10, which we carry over to the next place value; 9.0

7 Complete the table.

Number	Rounded to 1 d.p.	Rounded to 2 d.p.	Rounded to 3 d.p.
5.6	5.6	5.60	5.600
7.5	7.5	7.50	7.500
1.23	1.2	1.23	1.230
0.152	0.2	0.15	0.152
1.49	1.5	1.49	1.490
0.152	0.2	0.15	0.152
1.5015	1.5	1.50	1.502
1.2753	1.3	1.28	1.275
0.1204	0.1	0.12	0.120
2.3428	2.3	2.34	2.343
12.5007	12.5	12.50	12.501
1.6043	1.6	1.60	1.604
9.9899	10.0	9.99	9.990

8 Round these decimals as specified.

a 0.27 to the nearest tenth 0.3	b 125.91247 to the nearest tenth 125.9	c 321.4891 to the nearest hundredth 321.49
d 987.7983 to the nearest hundredth 987.80	e 0.73 to 1 decimal place 0.7	f 0.67321 to 3 decimal places 0.637
g 3.982178 to 4 decimal places 3.9822	h 2.987 to 1 decimal place 3.0	i 3.32913 to the nearest hundredth 3.33
j 19.9267 to the nearest tenth 19.9	k 0.2985 to 1 decimal place 0.3	l 0.2714 to 2 decimal places 0.27

9 Jamie goes to buy a watermelon. The clerk gives Jamie a watermelon that weighs 10.26 kg. What is the weight of the watermelon rounded to the nearest 100 g?

10.3 kg

10 James is ranked first on his cricket team with a batting average of 99.9387. Cricket batting averages are usually only calculated to two decimal places. What is James' batting average rounded to two decimal places?

99.94

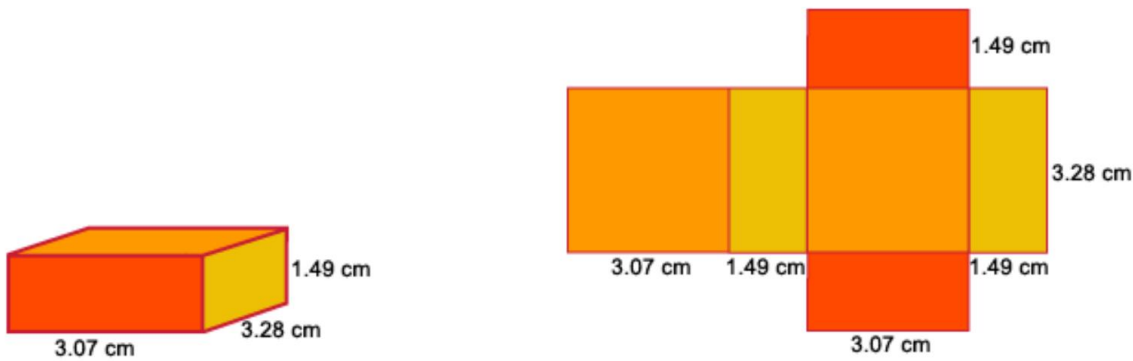
11 Samara believes 0.449999 should be rounded up to 0.5, but Cassandra believes it should be rounded down to 0.4. Who is correct, and why?

It should be rounded down to 0.4 as the critical digit in the hundredths place value is less than 5.

12 Josh is rounding 9.999 to two decimal places. He writes 9.90. Explain why he is incorrect.

When we round up from 9, it becomes 10, which we carry over to the next place value. This then causes the next place value to change.
The answer should be 10.00

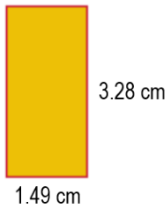
13 A common mistake students (even senior students!) make is rounding too early. We will find the area of the net of this box.



Rounding after 1 calculation:

Use a calculator to find the exact area of each rectangle, round each answer to 2 decimal places, then add them to find the total area of the net.

Left and right sides



Area of one side = 3.28×1.49
= 4.8872
 $\approx 4.89 \text{ cm}^2$

Area of both sides $\approx 9.78 \text{ cm}^2$

Top and bottom



Area of top =
=
 $\approx$

Area of top and bottom $\approx$

Front and back



Area of front =
=
 $\approx$

Area of bottom $\approx$

Total area of the net =

Rounding at the very end:

Use a calculator to find the exact area of each rectangle. Do not round at any stage.

Left and right sides



Area of one side =
=

Area of both sides =

Top and bottom



Area of top =

Area of top and bottom =

Front and back



Area of front =

Area of bottom =

Exact total area of the net =

Was there a difference between the two methods?

Terminating and Recurring Decimals

① Terminating and Recurring Decimals

Terminating decimals **end**: 0.25

Recurring decimals **repeat forever**: 0.3333333333333333333333333333...

Recurring digits are shown as:

- a dot for a single recurring digit. $0.666666 \dots = 0.\dot{6}$
- a bar for many recurring digits. $0.686868 \dots = 0.\overline{68}$

② Express recurring decimals using dot or bar notation

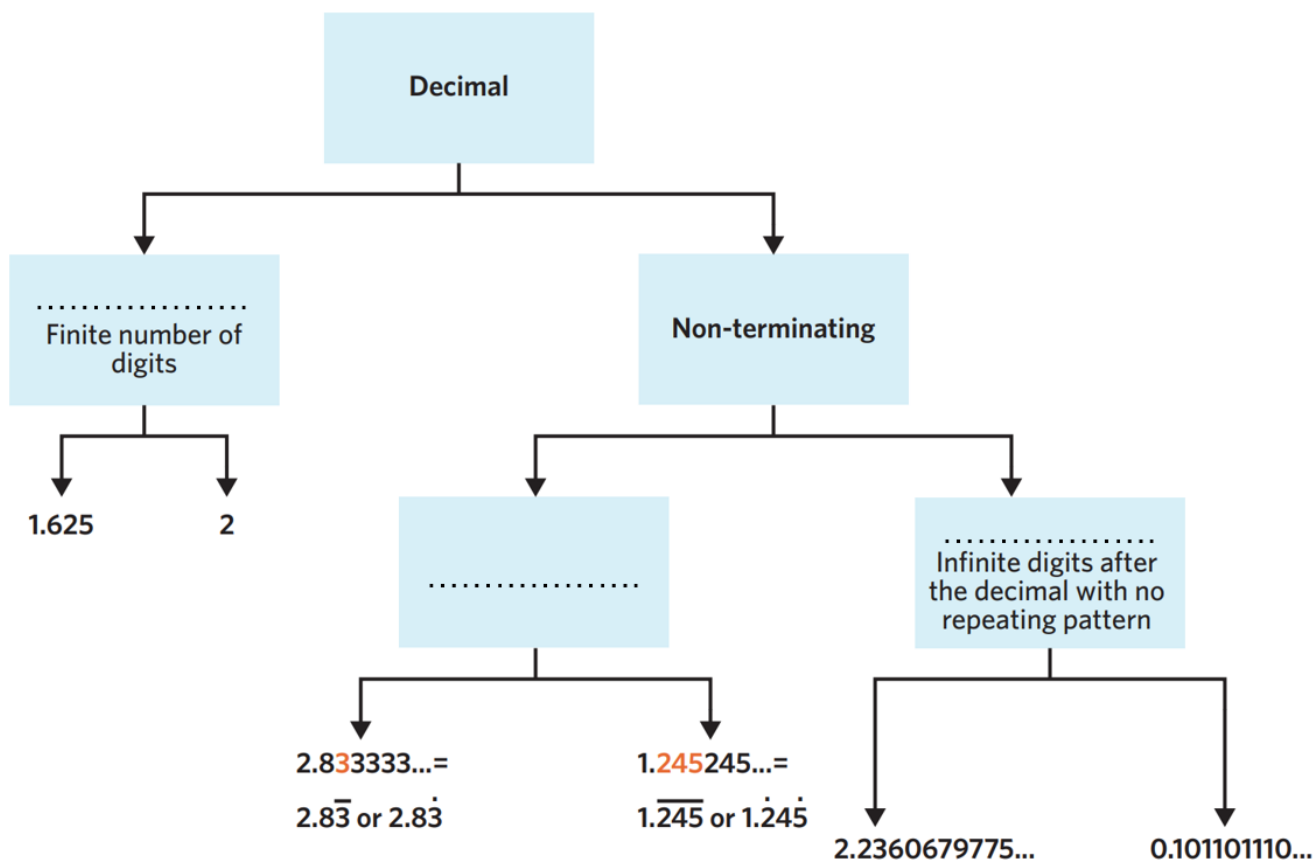
Express these recurring decimals using the appropriate notation

a 0.55555555555...	b 0.577777777777...	c 0.575757575757...
d 0.59595959595...	e 0.8214821482148214...	f 0.109421094210942...

Use a calculator to express these as decimals using the appropriate notation.

a $\frac{1}{3}$	b $\frac{2}{3}$	c $\frac{2}{7}$
d $\frac{5}{6}$	e $\frac{2}{15}$	f $\frac{18}{55}$

1 Fill in the missing words in the diagram.



From left to right: Terminating, Recurring, Non-Recurring

2 Classify these decimals as terminating, recurring, or non-recurring

	Terminating	Recurring	Non-recurring	
a 0.277777...	☐	☐	☐	recurring
b 6.40312436...	☐	☐	☐	non-recurring
c 0.234234234...	☐	☐	☐	recurring
d 1.625	☐	☐	☐	terminating

3 Classify these decimals as terminating or recurring.

a 0.5	b 0.555555555 ...	c 0.5	d 0.5555555555 ...
t	r	r	r
e 0.85	f 0.85	g 0.8585858585 ...	h 0.8585858585
t	r	r	t
i 0.76	j 0.325	k 0.1555555555 ...	l 0.00291
r	r	r	r

4 Write the following recurring decimals using the appropriate notation.

a	0.77777777...	b	0.35555555...	c	0.28282828...
	$0.\dot{7}$		$0.3\dot{5}$		$0.\overline{28}$
d	0.325325325...	e	0.678678678...	f	1.444444444...
	$0.\overline{325}$		$0.\overline{678}$		$1.\dot{4}$
g	6.922222222...	h	0.033333333...	i	0.909090909...
	$6.9\dot{2}$		$0.0\dot{3}$		$0.\overline{90}$
j	0.536666666...	k	3.217777777...	l	8.181818181...
	$0.53\dot{6}$		$3.21\dot{7}$		$8.\overline{18}$

5

a Use a calculator to convert $\frac{1}{3}$ to a recurring decimal and write using the correct notation.

$0.\dot{3}$

b Use a calculator to convert $\frac{2}{3}$ to a recurring decimal.
Explain why the calculator is not entirely correct.

The calculator says 0.666666667, it has rounded the answer when it ran out of digits.

c What would $1\frac{2}{3}$ be as a decimal?

$1.\dot{6}$

d What would $\frac{7}{3}$ be as a decimal?

$2.\dot{3}$

- 6 Use a calculator to determine whether these fractions represent terminating or recurring decimals. If they are recurring decimal, write using the correct notation.

a $\frac{4}{3}$	b $\frac{3}{3}$	c $\frac{1}{6}$	d $\frac{2}{6}$
1.3	1	0.1 $\overline{6}$	0.3
e $\frac{11}{30}$	f $\frac{18}{15}$	g $\frac{5}{9}$	h $\frac{7}{45}$
0.3 $\overline{6}$	1.2	0.5	0.1 $\overline{5}$
i $\frac{7}{8}$	j $\frac{4}{9}$	k $1\frac{5}{12}$	l $1\frac{2}{3}$
0.875	0.4	1.41 $\overline{6}$	1.6
m $\frac{11}{18}$	n $\frac{22}{33}$	o $1\frac{13}{22}$	p $\frac{11}{24}$
0.6 $\overline{1}$	0.6	0.59 $\overline{09}$	0.458 $\overline{3}$

- 7 The number π (pi) can be represented as a never-ending decimal. The first 100 digits of pi are:
 3.141 592 653 589 793 238 462 643 383 279 502 884 197 169 399 375 105 820 974 944 592
 307 816 406 286 208 998 628 034 825 342 117 067 ...

Would you classify π as a terminating or recurring decimal, or neither? Explain your reasoning.

Neither. It is a non-terminating, non-recurring decimal.